WACV 2027 UPAR Challenge

Fact sheet — Track 2

This fact sheet describes B137C, Codabench submission 1007281, the entry in the organisers' October 2026 leaderboard snapshot. It does not describe our later, unscored B153C candidate. This document is for challenge verification; we are not submitting a workshop paper. The section structure follows the UPAR fact-sheet template supplied by the organisers.

I. Team details

- Team leader name: Syouya Tobita.
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- Other team members and affiliations: None (individual participation).
- Team website URL: Not provided.
- Competition track: ( ) Track 1, Pedestrian Attribute Recognition; (X) Track 2, Attribute-based Person Retrieval.

II. Contribution details

A. Title of the contribution

Three-model attribute ensemble with spherical multi-resolution heads and gallery-only graph re-ranking.

We fine-tune three generic foundation models to predict 40 pedestrian attributes. Within each model family, same-seed recipe soups combine trained parameters without adding inference passes. Separate norm-preserving head soups combine attribute-attention and global evidence. Attribute-dependent member weights produce probabilities. An automatic, unlabelled-gallery re-ranker combines probability diffusion, visual anchors and distance smoothing. There is no cross-query aggregation or manual test-image intervention.

B. Representative image / workflow diagram

C. Detailed method description

1) Data and fit boundaries: We use the organiser-provided Track 2 annotations for Market1501, PA100k and PETA: 97,669 train images and 33,407 validation images. Final model recipes use train and validation together, excluding one exact-pixel duplicate on the training side, leaving 131,075 images. No extra pedestrian-labelled dataset or test images are used for model fitting. Generic pretraining is separate from this task-specific training.

Local model selection uses leave-one-domain-out predictions: each fold model is trained on the other sources, not the evaluated source. The pooled validation surface contains 33,407 images and 3,462 queries. These development surfaces and Public feedback were reused during selection; they are not untouched confirmation data. Local mADM and official mADM are not interchangeable.

2) Pretraining, input and architecture: The initialisations are PE-Core-L/14 [1], DINOv3-H+/16 [2], and EVA02-L/14 [3]. Their exact timm model identifiers are, respectively, vit_pe_core_large_patch14_336.fb, vit_huge_plus_patch16_dinov3.lvd1689m, and eva02_large_patch14_224.mim_m38m. PE uses generic image–text contrastive pretraining; DINOv3 uses self-supervised LVD-1689M pretraining; EVA02 uses masked image modelling on Merged-38M. Model cards and the bundled DINOv3 licence accompany the model ZIP.

Images are decoded to RGB and resized bilinearly to 256×128, then resized with antialiasing to each member's input and normalised with its recorded mean and standard deviation. Native-resolution detail is not preserved by this cache. There is no SAHI, sliding window, crop ensemble or flip at inference. PE has 316.1M parameters and 200 spatial patches; DINOv3-H+ has 840.9M parameters and 72 spatial patches (plus special tokens); EVA has 303.2M parameters and 162 spatial patches. PE and DINOv3 use a linear head plus a 40-query, 256-dimensional attribute-attention residual; EVA uses a linear head. All backbone layers are fine-tuned.

3) Training: There are 12 source training recipes, with seed 20261007: four PE, six DINOv3 and two EVA recipes. Each uses 400,000 source-balanced samples, effective batch size 128, AdamW, weight decay 0.05, 5% warm-up, cosine decay, head learning rate 10^{-3}, head dropout 0.1 and EMA 0.999. The first 10% of steps train the head only (LP-FT [4]). Backbone learning rate / layer-wise decay / drop path are $5 \cdot 10^{-5}/0.85/0.2$ for PE and EVA and $4 \cdot 10^{-5}/0.9/0.3$ for DINOv3. Training uses bfloat16 autocast.

The loss is positive-weighted binary cross-entropy. Some recipes add a listwise rank loss with weight 0.1; selected rank recipes also weight queries by inverse attribute-vector frequency. Other variants use per-source masks for missing/ambiguous annotations or lower-body part-swap augmentation. Common augmentations are

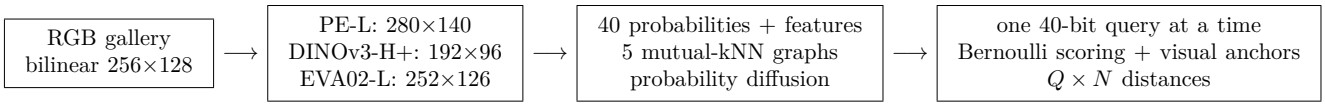

Fig. 1. Actual B137C inference structure. One forward pass per member and gallery image; no test-time augmentation. All graph statistics use only the unlabelled current gallery. Each query is processed without statistics from other queries. Dimensions are height by width.

horizontal flip, random affine, brightness and contrast jitter. The exact per-run JSON files, including accumulation and loss settings, are supplied. Prior correction subtracts the logarithm of the positive class weight before sigmoid at inference.

4) Parameter and head soups: Backbone soups [5] use weights 0.25 each for the four PE recipes; 0.20/0.30/0.30/0.20 for four 192-pixel DINOv3 recipes; and 0.5/0.5 for EVA recipes trained at 224×112 and 252×126. EVA positional embeddings are resampled to the target grid before mixing.

For a head weight row, let $u = \sum_i a_i w_i$. The spherical head uses $u(\sum_i a_i \|w_i\| / \max(\|u\|, 10^{-12}))^\tau$. PE head weights are 0.20/0.20/0.30/0.30 (base, rank, mask, part-swap), with positive-weight bias alignment for the mask variant, $\tau = 1.2$ for query/key, 1.0 for attention output and 0.6 for the linear base. DINOv3 head weights are 0.18/0.27/0.27/0.18 for the four 192-pixel recipes and 0.10 for their complementary 224-pixel head mixture; $\tau = 1$ and the attention output is scaled by 1.03. The complementary head is reconstructed in the original arithmetic order from the saved six-recipe soup and four 192-pixel heads. EVA uses a 0.5/0.5 head mix and $\tau = 1.2$. B137C does not use the later column-selective DINOv3 head.

Large two-dimensional tensors are stored with per-row int8 quantisation and half-precision scales; other tensors are half precision. The exact TorchScript runtime, weights and configuration are supplied in the original submission ZIP.

5) Scoring and re-ranking: The three probabilities are mixed per attribute (for example, 0.420/0.345/0.235 for age and gender in PE/DINOv3/EVA order). All 120 weights are fixed in assets/config.json; no dataset identifier is used at inference. Initial distances combine weighted absolute error and Bernoulli negative log-likelihood against each 40-bit query.

Five mutual-kNN graphs use PE, EVA, 0.75PE+0.25EVA, 0.5PE+0.5EVA and 0.5625PE+0.1875EVA+0.25DINOv3 feature similarities. Neighbourhood sizes are 20,20,28,28,28 and branch weights are 0.275,0.230,0.235,0.190,0.070. Edges combine visual similarity, predicted attribute agreement and closed-neighbourhood Jaccard overlap [6]. More than five confident attribute disagreements suppress an edge. Three probability diffusion steps are followed by graph re-estimation and six steps from the original probabilities, with confidence and member-disagreement weights.

Base, exact-match-emphasising and body scoring streams are smoothed on the graphs. Each query's top results provide visual anchors, with hubness normalisation and eight anchor similarity mixtures. Head/tail blending combines those streams. The B137C runtime has neither the later eight-step near-duplicate extension nor top-100 attribute pseudo-relevance feedback. Every constant and operation is preserved in the deployed b127_jaccard_prob_diffusion.py; that source, not later exploratory code or unused descriptive configuration, defines the entry.

6) Feedback use and compliance disclosure: The official scoring feedback for our submissions exposed per-dataset names, query/gallery counts and scores. During the project, these reports informed local proxy interpretation, emphasis on query match multiplicity, and synthetic-composition checks built solely from organiser training images. No hidden images or labels were accessed, and inference does not branch on dataset identity. This disclosure is project-wide and does not imply that every later analysis was used to select B137C. The challenge restricts use of explicit hidden-test structure. We ask the organisers to review whether this use of their scoring feedback is permitted; we do not claim that uniform inference alone settles that interpretation.

D. Challenge results

TABLE I

B137C, submission 1007281: exact values in the organisers' supplied leaderboard snapshot, matching the saved Finished receipt. Rank is snapshot rank, not a claim that winner verification has concluded.

Metric	Value
Rank in supplied snapshot	1 of 49
mADM	0.701395
mAP	0.418136
Rank-1	0.394486
Rank-5	0.567449
Rank-10	0.642728
mINP	0.293015

The saved B137C local OOF report records mADM 0.671415. Its official ingestion log reports approximately 258.3 seconds total against the 300-second limit, including approximately 18.4 seconds re-ranking. This is a completed run, not a worst-case hardware guarantee. A single three-member CUDA plan is required, with finite-value, canary and deadline checks; failure does not silently switch to a weaker model.

E. Final remarks

Soups and low-token inputs allow three large model families within the inference budget. Gallery context is especially helpful for queries with multiple relevant images. Limitations include information loss in resizing, repeated development-set selection, domain shift and a relatively tight runtime budget. The later B153C candidate is unscored and is not substituted for the entry reported here.

III. Additional method details

- Trained on predefined datasets? (X) Yes, ( ) No. Only organiser train/validation data for task-specific fitting.
- Pre-trained models? (X) Yes, ( ) No. The three generic checkpoints listed above, with original model cards and licence notices.
- External data? ( ) Yes, (X) No for task-specific training; generic external pretraining is explicitly declared above.
- Data augmentation? (X) Yes, ( ) No. Flip, affine, brightness/contrast; selected recipes also use part-swap.
- Validation data used in final training? (X) Yes, ( ) No.
- Regularisation? (X) Yes, ( ) No. Weight decay, dropout/drop path, LP-FT, EMA, source masks and parameter soups.
- Handcrafted features? ( ) Yes, (X) No. Visual features are learned; graph and distance operations are explicitly designed.
- Domain adaptation? ( ) Yes, (X) No parameter adaptation. Automatic unlabelled-gallery re-ranking is transductive inference, not test-time fitting.
- Prior to condition for visual variety? ( ) Yes, (X) No explicit test-domain prior. Training positive weights and their correction are stated.

IV. Code repository

Code, training configurations and reproduction instructions are supplied as B137C_verification_materials.zip, alongside the original model ZIP B137C_TRI_SPHERICAL_MULTIRES_HEAD_RERANK.zip. No public repository is required to use these local files. Transfer to the organisers will be private and arranged separately; this document itself is not evidence of delivery. The model ZIP SHA-256 is bae4b4c90846bf243d79bf65b5e0929dba5b0725903731c50fea264a05a83077.

Inference reproduction uses the exact submitted bytes. Training code and 12 original recipes are included, with a helper that rewrites only cache and output paths into a new directory. A contemporary source snapshot and recorded environments are supplied; a clean environment recreation and bit-exact retraining are not claimed. Some original intermediate checkpoints and training-time source hashes were not retained. The trained submission weights remain intact. The six-recipe head mixture is preserved in a compact head-only checkpoint; this does not recreate its deleted parent checkpoints. The reproduction guide distinguishes inference, recipe retraining and historical provenance limitations rather than treating them as equivalent.

References

[1] D. Bolya et al., "Perception encoder: The best visual embeddings are not at the output of the network," arXiv preprint arXiv:2504.13181, 2025.

[2] O. Siméoni et al., "DINOv3," arXiv preprint arXiv:2508.10104, 2025.

[3] Y. Fang, Q. Sun, X. Wang, T. Huang, X. Wang, and Y. Cao, "EVA-02: A visual representation for neon genesis," arXiv preprint arXiv:2303.11331, 2023.

[4] A. Kumar, A. Raghunathan, R. Jones, T. Ma, and P. Liang, "Fine-tuning can distort pretrained features and underperform out-of-distribution," in International Conference on Learning Representations (ICLR), 2022.

[5] M. Wortsman, G. Ilharco, S. Y. Gadre, R. Roelofs, R. Gontijo-Lopes, A. S. Morcos, H. Namkoong, A. Farhadi, Y. Carmon, S. Kornblith, and L. Schmidt, "Model soups: averaging weights of multiple fine-tuned models improves accuracy without increasing inference time," in International Conference on Machine Learning (ICML), 2022.

[6] Z. Zhong, L. Zheng, D. Cao, and S. Li, "Re-ranking person re-identification with k-reciprocal encoding," in IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2017.